

CPT204 Data Structures and Algorithms - 22-23R

Part B. Coding Questions

Question 1: Coding 1

Complete the overridden method void addLast(T item).

It is a method of the class DelimitedARDeque, a subclass of an iterable ARDeque.

It adds an item at the back of the object of DelimitedARDeque,

but it will throw an IllegalArgumentException if the item already occurs in it more than once.

Test case:

```
DelimitedARDeque<String> dd = new DelimitedARDeque<>();  
dd.addLast("a");  
dd.addLast("b");  
dd.addLast("a");  
dd.printDeque();
```

Question 2: Coding 2

Complete the method boolean equals(Object that) of an iterable ARDeque that overrides the one in Object. To be the same, they must be of the same size, and must share at most 2 different items at the same position. Suppose [] is an ARDeque. Then: [1, 2, 3, 4].equals([1, 2, 5, 6]) -> true [1, 2, 3, 4].equals([1, 4, 2, 3]) -> ohf false